

Unit 16: Linearization and Differentials — Full Solutions

Warm-up

1. $f(x) = x^2$ at $a = 3$.

$f(3) = 9$. $f'(x) = 2x$, so $f'(3) = 6$.

$$L(x) = 9 + 6(x - 3)$$

Estimate $f(3.1)$: $L(3.1) = 9 + 6(0.1) = 9 + 0.6 = 9.6$.

$L(x) = 9 + 6(x - 3)$, and $f(3.1) \approx 9.6$. (The true value is 9.61 — very close.)

2. $f(x) = x^3$ at $a = 2$.

$f(2) = 8$. $f'(x) = 3x^2$, so $f'(2) = 12$.

$$L(x) = 8 + 12(x - 2)$$

Estimate $f(2.05)$: $L(2.05) = 8 + 12(0.05) = 8 + 0.6 = 8.6$.

$L(x) = 8 + 12(x - 2)$, and $f(2.05) \approx 8.6$. (The true value is 8.615125.)

3. $f(x) = \frac{1}{x}$ at $a = 4$.

$f(4) = 0.25$. $f'(x) = -\frac{1}{x^2}$, so $f'(4) = -\frac{1}{16} = -0.0625$.

$$L(x) = 0.25 - 0.0625(x - 4)$$

Estimate $f(4.2)$: $L(4.2) = 0.25 - 0.0625(0.2) = 0.25 - 0.0125 = 0.2375$.

$L(x) = 0.25 - 0.0625(x - 4)$, and $f(4.2) \approx 0.2375$. (The true value is ≈ 0.2381 .)

4. $f(x) = \sqrt{x}$ at $a = 9$.

$f(9) = 3$. $f'(x) = \frac{1}{2\sqrt{x}}$, so $f'(9) = \frac{1}{6}$.

$$L(x) = 3 + \frac{1}{6}(x - 9)$$

Estimate $f(9.3)$: $L(9.3) = 3 + \frac{1}{6}(0.3) = 3 + 0.05 = 3.05$.

$L(x) = 3 + \frac{1}{6}(x - 9)$, and $\sqrt{9.3} \approx 3.05$. (The true value is ≈ 3.0496 .)

5. $f(x) = x^2 + 3x$ at $a = 1$.

$f(1) = 1 + 3 = 4$. $f'(x) = 2x + 3$, so $f'(1) = 5$.

$$L(x) = 4 + 5(x - 1)$$

Estimate $f(1.1)$: $L(1.1) = 4 + 5(0.1) = 4 + 0.5 = 4.5$.

$L(x) = 4 + 5(x - 1)$, and $f(1.1) \approx 4.5$. (The true value is 4.51.)

6. $f(x) = \sqrt{x}$ at $a = 25$.

$$f(25) = 5. \quad f'(25) = \frac{1}{2\sqrt{25}} = \frac{1}{10} = 0.1.$$

$$L(x) = 5 + 0.1(x - 25)$$

Estimate $f(24.7)$: here $dx = 24.7 - 25 = -0.3$.

$$L(24.7) = 5 + 0.1(-0.3) = 5 - 0.03 = 4.97$$

$L(x) = 5 + 0.1(x - 25)$, and $\sqrt{24.7} \approx 4.97$. (The true value is ≈ 4.9699 .)

Standard

7. $y = x^3$, $x = 2$, $dx = 0.1$.

$$dy = f'(x) dx = 3x^2 dx$$

$$dy = 3(2)^2(0.1) = 3(4)(0.1) = 1.2$$

$$dy = 1.2.$$

8. $y = \sqrt{x}$, $x = 16$, $dx = -0.2$.

$$dy = \frac{1}{2\sqrt{x}} dx$$

$$dy = \frac{1}{2\sqrt{16}}(-0.2) = \frac{1}{8}(-0.2) = -0.025$$

$$dy = -0.025.$$

9. Estimate $\sqrt{50}$ using $a = 49$.

$$f(x) = \sqrt{x}. \quad f(49) = 7. \quad f'(49) = \frac{1}{2\sqrt{49}} = \frac{1}{14}.$$

$$dx = 50 - 49 = 1.$$

$$L(50) = 7 + \frac{1}{14}(1) = 7 + \frac{1}{14} \approx 7 + 0.0714 = 7.0714$$

$\sqrt{50} \approx 7.0714$. (The true value is ≈ 7.0711 .)

10. Estimate $\sqrt[3]{26}$ using $a = 27$.

$$f(x) = x^{1/3}. \quad f(27) = 3. \quad f'(x) = \frac{1}{3}x^{-2/3}, \text{ so } f'(27) = \frac{1}{3}(27)^{-2/3} = \frac{1}{3} \cdot \frac{1}{9} = \frac{1}{27}.$$

$$dx = 26 - 27 = -1.$$

$$L(26) = 3 + \frac{1}{27}(-1) = 3 - \frac{1}{27} \approx 3 - 0.037 = 2.963$$

$\sqrt[3]{26} \approx 2.963$. (The true value is ≈ 2.9625 .)

11. $y = \frac{1}{x^2} = x^{-2}$, $x = 5$, $dx = 0.05$.

$$f'(x) = -2x^{-3} = -\frac{2}{x^3}$$

$$dy = -\frac{2}{5^3}(0.05) = -\frac{2}{125}(0.05) = -0.016(0.05) = -0.0008$$

$$dy = -0.0008.$$

12. Estimate $(3.98)^2$ using $a = 4$.

$$f(x) = x^2. \quad f(4) = 16. \quad f'(4) = 8.$$

$$dx = 3.98 - 4 = -0.02.$$

$$L(3.98) = 16 + 8(-0.02) = 16 - 0.16 = 15.84$$

$(3.98)^2 \approx 15.84$. (The true value is 15.8404 — an excellent match.)

Challenge

13. Sphere, $V = \frac{4}{3}\pi r^3$, $r = 10$, $dr = \pm 0.1$.

$$dV = 4\pi r^2 dr$$

$$dV = 4\pi(10)^2(0.1) = 4\pi(100)(0.1) = 40\pi \approx 125.66 \text{ cm}^3$$

The estimated possible error in the computed volume is about $40\pi \approx 125.7 \text{ cm}^3$.

14. Circle, $A = \pi r^2$, $r = 5$, $dr = \pm 0.05$.

$$dA = 2\pi r dr = 2\pi(5)(0.05) = 0.5\pi \approx 1.5708 \text{ cm}^2$$

Percentage error: divide dA by the actual area $A = \pi(5)^2 = 25\pi$, then multiply by 100%:

$$\frac{dA}{A} \times 100\% = \frac{0.5\pi}{25\pi} \times 100\% = \frac{0.5}{25} \times 100\% = 2\%$$

The estimated error in the computed area is about 1.57 cm², or approximately 2%.

15. $f(x) = \tan x$ at $a = \frac{\pi}{4}$.

$$f\left(\frac{\pi}{4}\right) = \tan\left(\frac{\pi}{4}\right) = 1.$$

$$f'(x) = \sec^2 x. \text{ At } x = \frac{\pi}{4}: \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}, \text{ so } \cos^2\left(\frac{\pi}{4}\right) = \frac{1}{2}, \text{ and } \sec^2\left(\frac{\pi}{4}\right) = \frac{1}{1/2} = 2.$$

$$L(x) = 1 + 2\left(x - \frac{\pi}{4}\right)$$

Estimate $f\left(\frac{\pi}{4} + 0.01\right)$: here $dx = 0.01$.

$$L\left(\frac{\pi}{4} + 0.01\right) = 1 + 2(0.01) = 1.02$$

$$\tan\left(\frac{\pi}{4} + 0.01\right) \approx 1.02. \text{ (The true value is } \approx 1.0203\text{.)}$$

16. $f(x) = \sin x$ at $a = 0$.

$$f(0) = 0. \quad f'(x) = \cos x, \text{ so } f'(0) = 1.$$

$$L(x) = 0 + 1(x - 0) = x$$

Estimate $\sin(0.05)$: $L(0.05) = 0.05$.

$\sin(0.05) \approx 0.05$. (The true value is ≈ 0.049979 — remarkably close. This is the classic “small angle approximation” $\sin x \approx x$ for small x measured in radians.)

17. Cube, $V = s^3$, $s = 6$, $ds = \pm 0.05$.

$$dV = 3s^2 ds = 3(6)^2(0.05) = 3(36)(0.05) = 5.4 \text{ cm}^3$$

Percentage error: the actual volume is $V = 6^3 = 216 \text{ cm}^3$.

$$\frac{dV}{V} \times 100\% = \frac{5.4}{216} \times 100\% = 2.5\%$$

The estimated error in the computed volume is about 5.4 cm³, or approximately 2.5%.

Applied

18. $C(x) = 0.01x^2 + 5x + 200$.

$$C'(x) = 0.02x + 5$$

At $x = 100$: $C'(100) = 0.02(100) + 5 = 2 + 5 = 7$.

Using the differential with $dx = 1$:

$$dC = 7(1) = 7$$

Estimated change: \$7.

Now the exact change: $C(100) = 0.01(10000) + 500 + 200 = 100 + 500 + 200 = 800$. $C(101) = 0.01(10201) + 505 + 200 = 102.01 + 505 + 200 = 807.01$.

$$\Delta C = 807.01 - 800 = 7.01$$

Comparison: the differential estimate (\$7.00) is extremely close to the exact change (\$7.01).

19. $P(t) = 500 + 20t + t^2$.

$$P'(t) = 20 + 2t$$

At $t = 10$: $P'(10) = 20 + 20 = 40$.

Using $dt = 0.2$:

$$dP = 40(0.2) = 8$$

Estimated change in population: about 8 individuals.

20. $h(t) = -16t^2 + 200t + 10$.

$$h'(t) = -32t + 200$$

At $t = 3$: $h'(3) = -96 + 200 = 104$.

Using $dt = 0.1$:

$$dh = 104(0.1) = 10.4$$

Estimated change in height: about 10.4 feet.

21. $R(x) = 100x - 0.5x^2$.

$$R'(x) = 100 - x$$

At $x = 50$: $R'(50) = 100 - 50 = 50$.

Using $dx = 1$:

$$dR = 50(1) = 50$$

Estimated change: \$50.

Now the exact change: $R(50) = 100(50) - 0.5(2500) = 5000 - 1250 = 3750$. $R(51) = 100(51) - 0.5(2601) = 5100 - 1300.5 = 3799.5$.

$$\Delta R = 3799.5 - 3750 = 49.5$$

Comparison: the differential estimate (\$50.00) is very close to the exact change (\$49.50) — the small gap is expected since $dx = 1$ isn't infinitesimally small, but the estimate is still quite good.